

Which Head Does the Work? How Attention-Head Circuit Findings Carry Over Between Related Language Models

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Abstract

Circuit analyses name the attention heads that carry a computation, and these names are routinely reused: on later checkpoints, on fine-tuned descendants, and on sibling models trained from the same initialization on other data. We ask in which related models the named heads actually do the work. Using public model suites that cross initialization with training data (69 DataDecide models at 1B parameters on 25 corpora; Pythia 160M–12B and its deduplicated sibling), the full OLMo 2 lineage from pretraining through midtraining, SFT, DPO and instruction tuning, and controlled pretraining, we separate three things that a head-level finding can mean. The layer that hosts a role and the algorithm a circuit runs are shared by all models. The attention role of an individual head is shared among models trained from the same initialization, across all 25 corpora, and not between models from different initializations. The causal carrier of a task is assigned by training: siblings that share the initialization and differ only slightly in data run the same indirect-object-identification algorithm on different heads, while a model’s own carriers, once settled in pretraining, stay in place through midtraining, preference tuning and instruction tuning. Controlled pretraining exposes the mechanism: candidate heads compete for a role while it forms, and when two candidates are close, masking a random 10% of the data hands the role to the other one with the ability intact. Head-level findings therefore carry over along a training lineage, and attention-role labels carry over within an initialization; across separately trained runs, the algorithm carries over and the carriers are found anew.

1 Introduction

Mechanistic interpretability explains a computation by locating the components that carry it. A circuit for indirect-object identification (IOI) is a set of

named heads, *e.g.*, the name movers and the heads that suppress the repeated name (Wang et al., 2023); an induction circuit is a previous-token head composed with an induction head (Olsson et al., 2022). Such findings are then used beyond the model in which they were found. Circuits are tracked across the checkpoints of a training run (Tigges et al., 2024), compared between a base model and its fine-tuned versions (Prakash et al., 2024), and used as the target of training-data interventions that retrain a model and read out the same head again (Chen et al., 2026). Each of these uses assumes that a specific head in one model corresponds to a specific head in another.

Which correspondences hold is not determined by the architecture. Heads within a layer are interchangeable: permuting them, together with their output projections, leaves the function unchanged, so a head’s index carries meaning only relative to a particular training history. Prior work has shown that circuits are consistent in their algorithm across training and scale while their components change (Tigges et al., 2024), that individual heads are unstable across random seeds (Bali et al., 2026), and that positional agreement across Pythia seeds is limited (Trott, 2025). These studies vary the seed, which changes the initialization and the data order together, and they compare runs that were trained separately. The uses listed above, however, involve two other relations between models: siblings that share an initialization but see different data, and descendants that continue the training of the same model.

We separate the relations with populations in which they can be varied one at a time (§2). DataDecide crosses three initializations with 25 pretraining corpora at 1B parameters (Magnusson et al., 2025); Pythia releases a standard and a deduplicated model from the same initialization at every size, with intermediate checkpoints (Biderman et al., 2023); OLMo 2 releases every stage of

one lineage from pretraining to instruction tuning (Team OLMo et al., 2024); and controlled pretraining lets us intervene on the data of a single run. In each model we measure attention roles of every head and the causal carriers of IOI, with split-half reliability for every head map.

Three findings organise the paper. (1) **Initialization fixes a shared coordinate system (§3)**. All models agree on the layer that hosts each of nine head roles; which head within the layer takes the role is shared by all models from the same initialization across 25 corpora and by no other model. (2) **Training assigns the causal carriers within that coordinate system (§4)**. Siblings from the same initialization run the same IOI algorithm with different heads, including siblings whose corpora differ only by the removal of 1.5% of the data, and, at four of six Pythia sizes, by deduplication. (3) **Carriers move during pretraining and then stay (§5)**. Along one run, responsibility moves between heads through most of pretraining, and the carriers present at the end of pretraining are the carriers after midtraining, SFT, DPO and instruction tuning. Controlled pretraining shows the competition directly (§6): when one induction head leads, masking training data during its emergence delays that head; when two heads are nearly equal, masking a random 10% of the data hands the role to the other head with the ability intact.

These findings give a practical rule (§8): a head-level finding carries over along a training lineage; an attention-role label carries over within an initialization; across separately trained runs the algorithm carries over and the carriers are localized again. Along the way, the head layouts identify four public checkpoints whose lineage differs from their labels, including the released OLMo-2-1B base model (Appendix A).

2 Setup

Model populations. *DataDecide-1B* trains three initializations (seeds default, large-aux-2, large-aux-3) on 25 corpora for 100B tokens (Magnusson et al., 2025). We verified every run’s initialization against the released step-0 weights and use the 69 runs whose initialization matches its label (Appendix A). Pairs of runs are either *siblings* (same initialization, different corpus) or *unrelated* (different initialization). *Pythia* trains a standard and a deduplicated model from the same initialization at each size (Biderman et al., 2023); we

Population	Models	Relation varied	Used in
DataDecide-1B	25 corpora $\times$ 3 init.; 69 with verified init.	corpus, init.	§3, §4, §5
DataDecide sizes	4M–750M, 10–23 corpora	corpus, init., scale	§3
Pythia	70M–12B, standard and deduplicated	data (same init.)	§3, §4
Pythia checkpoints	160M, 410M pairs, 15 steps each	training time	§5
PolyPythias	410M, seed 1	init. and data order	§4
OLMo-2-1B	pretraining end, 3 mid-training runs, SFT, DPO, Instruct	continued training	§5
Controlled	4- and 12-layer, C4, 2–3 init.	perturbations, data masking	§3, §6

Table 1: Model populations and the relation between models that each one varies.

use 160M to 12B and fifteen checkpoints of the 160M and 410M pairs. *OLMo-2-0425-1B* provides one lineage: the end of pretraining (4T tokens), three midtraining runs on different mixtures (51B tokens each), and the SFT, DPO and Instruct models (Team OLMo et al., 2024). *Controlled pretraining* trains 12-layer, 12-head Llama-style models on C4 from three initializations with a fixed data order (Appendix C).

Table 1 summarizes the populations and the relation that each one varies.

Attention roles. For every head we compute nine role scores (Olsson et al., 2022; Xiao et al., 2024; Wu et al., 2025): induction, previous token, attention sink, knowledge retrieval, duplicate token, current token, two tokens back, delimiter, and a weight-only OV copying score (Appendix B). Each score gives a [layer $\times$ head] map per model.

Causal carriers of IOI. We use the IOI dataset of Wang et al. (2023) (200 prompts, mixed templates) and, for every head, (i) its direct logit attribution to the IO name over the subject name at the final position, with every normalization linearized at its actual scale, (ii) the drop in logit difference when the head’s input to the output projection is mean-ablated at the final position, and (iii) its attention from the final position to the IO and S2 tokens. The carriers of a model are its heads with the largest direct effect; we compare carriers between models by the overlap of their top three heads and by *transfer*, the damage that ablating one model’s top-three heads does in the other model relative to the other model’s own top three.

Comparing head maps. Two models’ maps are compared within each layer by rank correlation, averaged over layers, so that agreement on which layer hosts a role does not count as agreement on which head does. Every map is computed on two

	induction	END→IO	ablation	DLA (rank)	DLA (value)
DataDecide-1B (9 models)	1.00	0.99	0.83	0.50	0.98
Pythia-410M	0.99	0.99	0.83	0.40	0.99
OLMo-2-1B (pretraining end)	1.00	0.99	0.86	0.42	0.98

Table 2: Split-half reliability of head maps (Spearman–Brown corrected). Rank statistics are reliable for attention maps and ablation maps; for direct logit attribution (DLA) the values of the contributing heads are reliable, while the ranks of the many near-zero heads are not.

Role	which head within the layer			which layer
	same init. diff. corpus	diff. init. same corpus	both differ	same init. / both differ
Induction	0.33	-0.02	0.00	0.49 / 0.41
Previous token	0.42	-0.01	-0.00	0.46 / 0.38
Attention sink	0.18	-0.01	-0.00	0.64 / 0.58
Knowledge retrieval	0.31	-0.00	0.00	0.60 / 0.54
Duplicate token	0.36	-0.02	-0.01	0.62 / 0.57
Current token	0.32	-0.02	-0.00	0.61 / 0.59
Two tokens back	0.43	-0.00	0.01	0.62 / 0.58
Delimiter	0.22	-0.01	0.00	0.57 / 0.50
OV copying (weights only)	0.25	-0.01	-0.01	0.68 / 0.64

Table 3: Correspondence of head-role maps across the 69 DataDecide-1B models. Which head within a layer takes a role is shared by siblings (same initialization, different corpus) and by no other pair; the profile over layers is shared by every pair.

halves of its inputs, and correspondences between models are divided by the geometric mean of the two models’ split-half reliabilities. This matters for causal maps: most heads contribute nothing to IOI, so a rank correlation over all heads in a layer is dominated by the order of near-zero values and reaches a split-half reliability of only 0.40–0.50 for direct logit attribution, although the values of the contributing heads replicate almost perfectly (0.98). Single-head ablation maps are reliable under the rank statistic (0.83–0.86) and are our causal map; carrier identity and transfer use the contributing heads directly (Table 2).

3 Initialization fixes a coordinate system

Across the 69 DataDecide-1B models, the layer profile of every role is shared by all pairs of models, related or not, and which head inside the layer takes the role follows the initialization (Figure 1, Table 3). Ordered by initialization, the pairwise correspondences form three blocks: siblings correspond at 0.31 on average and models from different initializations at 0.00, and the two distributions barely touch (lowest sibling pair 0.10, highest unrelated pair 0.14). Siblings trained on different corpora agree on the head for every one of the nine roles (within-layer correspondence 0.18–0.43), whereas models from different initializations agree

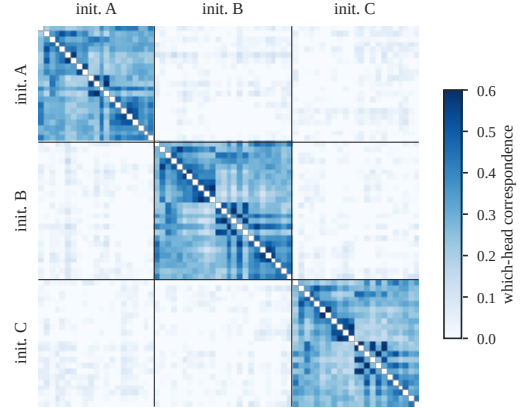


Figure 1: Which-head correspondence between all pairs of the 69 DataDecide-1B models (mean within-layer correspondence of four attention roles), ordered by initialization and, within each block, by corpus. Models from one initialization form one block across all 25 corpora.

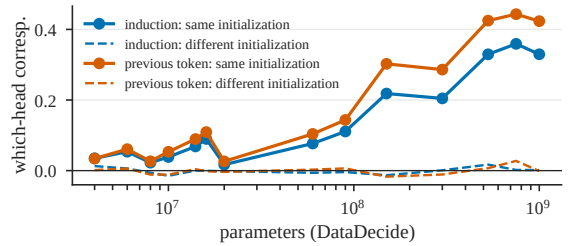


Figure 2: The shared coordinate system across 14 DataDecide sizes: within-layer correspondence of induction and previous-token maps between siblings (solid) and between models from different initializations (dashed).

at chance whether they share the corpus or not. A per-head variance decomposition makes the same point: the initialization has a significant effect on 66–79% of heads for the attention roles, the corpus on 4–7%, the rate expected by chance. The weight-only OV copying score behaves like the attention roles, so the shared assignment is a property of the weights and not of the probe inputs.

Across scale and model families. The coordinate system strengthens with scale (Figure 2). Across the 14 DataDecide sizes, sibling correspondence is small below 20M parameters, clear from 60M, and reaches 0.33–0.44 at 0.5–1B, while models from different initializations stay at zero at every size. The second family behaves the same way: the standard and deduplicated Pythia models, which share their initialization and differ in data,

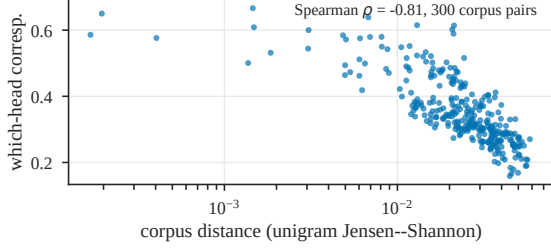


Figure 3: DataDecide-1B siblings share more of the coordinate system when their corpora are closer: which-head correspondence (mean over induction, previous-token and retrieval maps and over the three initializations) against the unigram Jensen-Shannon distance between the two corpora.

correspond at 0.38–0.72 on the induction, previous-token and retrieval maps at all six sizes we compare between 70M and 12B (Appendix A explains why 2.8B is left out), against a 95th percentile of 0.04–0.26 for head permutations.

Set in the first percent of training. Controlled pretraining shows when the coordinate system is fixed (Table 4). Adding noise as large as the weights themselves at 1% of training rearranges which heads take the roles (correspondence with the uninterrupted run 0.36); the same perturbation from 2.5% on is absorbed, and the heads return to their slots (0.85–0.98, against 0.97 for an exact rerun). A switch of the training corpus to code after this point keeps a correspondence of 0.60–0.95 with the uninterrupted layout, higher the later it happens. The public suites agree: DataDecide-1B siblings share the coordinate system at 3.6% of training as strongly as at the end.

How much siblings share. Siblings share part of the coordinate system, and the part grows with the similarity of their data (Figure 3). Over the 300 pairs of DataDecide corpora, the closer two corpora are in their word distribution, the more of the layout their siblings share (Spearman $\rho = -0.81$), from 0.16 for the most distant corpora to 0.67 for the closest.

The initialization therefore defines a coordinate system that all its descendants share: in two siblings, “layer 12, head 3” refers to a head with the same attention role, and component-level comparisons between siblings need no alignment. This holds across all 25 corpora, which range from web crawls to curated and quality-filtered mixtures.

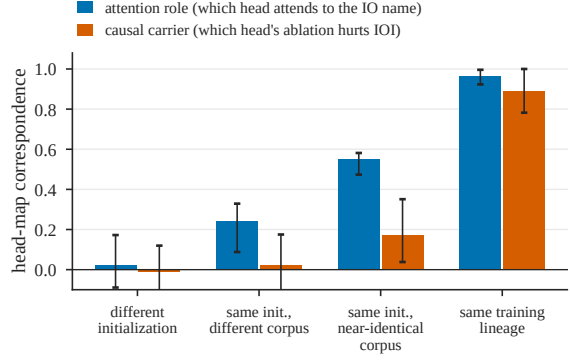


Figure 4: What carries over between two models, by their relation. Blue: correspondence of the attention map from the final position to the IO name; orange: reliability-corrected correspondence of the single-head ablation map for IOI. Bars are means over model pairs, whiskers the range. Siblings share the attention role of a head but not its causal role; models in one training lineage share both.

4 Training assigns the carriers

Within the shared coordinate system, a task’s causal carriers are a separate matter (Figure 4). For IOI, the attention role of a head, whether it attends from the final position to the IO name, corresponds between DataDecide siblings at 0.24, the same level as the roles of §3. The causal map does not: its reliability-corrected correspondence between siblings is 0.02, as low as between models with different initializations (0.01), and the top-three carriers of one sibling transfer 3% of the damage to another. Of the 18 sibling pairs among four DataDecide corpora, none share the main carrier. Comparing the values of the direct logit attribution instead of their ranks gives the same picture: siblings correspond at 0.11 and models from different initializations at 0.07–0.10. The same contrast holds over all 69 DataDecide-1B models: siblings share the END→IO attention map at 0.29 and models from different initializations at 0.00, while the single-head ablation map of IOI corresponds at 0.07 between siblings and at 0.00–0.01 otherwise, and a sibling’s top three to top twenty heads transfer 7–16% of the damage, against 4–12% for unrelated models. Across these sibling pairs, carrier transfer barely depends on how close the two corpora are (Spearman $\rho = -0.10$ with corpus distance), whereas the shared coordinate system follows corpus distance closely ($\rho = -0.81$, §3).

The assignment is equally independent of how

Step	Share	300-step warm-up				1,000-step warm-up		Optimizer reset
		Switch to code	Noise 100%	Noise 10%	Code, 12 layers	Switch to code	Noise 100%	Noise 100%
0	0%	−0.03	0.03	0.74	−0.03	—	—	—
100	1%	0.09	0.36	0.94	0.40	—	0.16	—
250	2.5%	0.60	0.85	0.97	—	0.41	0.70	0.81
500	5%	0.72	0.96	0.99	—	—	0.87	0.91
1000	10%	0.71	0.95	0.99	0.84	0.69	0.87	0.97
2000	20%	0.81	0.98	1.00	—	—	0.94	—
4000	40%	0.86	0.96	1.00	—	—	—	—
8000	80%	0.95	0.98	1.00	—	—	—	—
exact rerun					0.97			

Table 4: When the coordinate system is set (controlled 4-layer models, means over two initializations; one 12-layer model for the code switch in the fourth column; Appendix C). Which-head correspondence (mean of induction and previous-token maps) between the final layout of a run that was intervened on at a given step and the uninterrupted run: switching the corpus to code, or adding Gaussian noise scaled to 100% or 10% of each tensor’s standard deviation, under the default 300-step warm-up, a 1,000-step warm-up, and with a freshly initialized optimizer. Step 0 for the code switch: the same initialization trained on code from scratch.

Size	main carrier (standard)	also in sibling’s top three	top-3 shared	transfer
160M	8.9	yes	2/3	0.85
410M	11.4	no	0/3	0.05
1B	12.1	no	0/3	0.90
1.4B	10.7	no	0/3	−0.05
6.9B	20.28	no	0/3	0.04
12B	20.11	no	0/3	0.16

Table 5: IOI carriers of standard and deduplicated Pythia models (same initialization). Transfer: damage of the standard model’s top three heads in the deduplicated model, relative to the latter’s own top three.

Model	subject-suppression head				name movers	negative name movers
	head	effect	attn. to S2	OV copy		
Pythia-160M	8.9	1.43	0.94	−1.52	4	1
Pythia-160M-dedup.	8.9	1.65	0.90	−1.36	3	2
Pythia-410M	11.4	1.11	0.39	−1.81	14	2
Pythia-410M-dedup.	12.12	0.95	0.34	−1.83	11	3
Pythia-410M seed 1	12.2	2.28	0.54	−1.99	12	5

Table 6: The IOI algorithm in five Pythia models. Every model has a subject-suppression head, which attends to the repeated subject (S2) and copies it negatively (OV copy score < 0), and several name movers and negative name movers; the head that plays the suppression role differs between siblings at 410M. Effect: direct effect on the logit difference on held-out prompts.

similar the data are. Removing the Flan instruction data, about 1.5% of the Dolma corpus, gives three sibling pairs whose attention roles agree most closely of all the sibling pairs we compare (0.57–0.58), and each pair carries IOI on different main heads. The standard and deduplicated Pythia models carry IOI on largely the same heads at 160M and 1B (transfer 0.85 and 0.90) and on different heads at 410M, 1.4B, 6.9B and 12B (transfer between −0.05 and 0.16; Table 5); at 410M their attention roles still correspond at 0.48 while their causal maps correspond at 0.15.

What differs is the head, not the computation. In the Pythia models we analysed head by head (160M and 410M, standard and deduplicated, and an independently initialized 410M), IOI runs on the same functional classes: a head that attends to the repeated subject and writes against it through a negative OV copying circuit, and several name movers that attend to the IO name and copy it (Wang et al., 2023). Each class has several members in each

model, and siblings give the main role to different members (Table 6, Figure 5).

In the 410M Pythia pair, the strongest carrier of the standard model, 11.4, contributes 1.16 to the logit difference and its counterpart in the deduplicated model 0.02, while the deduplicated model’s strongest carrier, 12.12, contributes 0.97 and its counterpart 0.00; each is nearly silent in the other model, attending to the first token instead.

Because several heads in a class share the attention profile, attention and OV profiles do not determine which member carries the task (Table 7). Mapping one model’s carriers onto another by the same index, by the nearest attention and OV profile, or by a class rule recovers the target’s carriers for some pairs and fails for others; the mapping succeeds for every pair as soon as it includes a causal readout in the target (85–94% of the target’s own

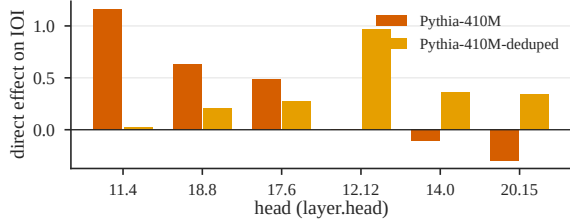


Figure 5: Same algorithm, different carriers. Direct effect on the IOI logit difference of each sibling’s top-three heads, measured in both siblings (Pythia-410M and its deduplicated version, same initialization). Each model’s main carrier is silent in the other.

Mapping	same initialization		different initialization 410M seed 1
	160M-dedup.	410M-dedup.	
Same head index	0.87	0.06	0.00
Nearest attention / OV profile	0.92	0.84	−0.06
Class rule (attention and OV sign)	0.92	0.18	1.06
Profile with the target’s causal readout	0.89	0.85	0.94

Table 7: Mapping the standard Pythia model’s top-three IOI heads onto a target model: damage of the mapped heads in the target relative to the target’s own top three, on held-out prompts. Only a mapping that uses a causal readout in the target recovers the carriers for every pair.

top-three damage). The carrier has to be identified causally in each model.

5 Carriers move during pretraining, then stay

Descendants of one model keep its carriers (Table 8). From the end of OLMo-2 pretraining to the end of each of three midtraining runs on different mixtures, the causal maps correspond at 0.84–0.88, two of the three top carriers are retained, and the earlier carriers produce 94–95% of the later model’s own damage; the main carrier, 14.4, is the same throughout. SFT, DPO and instruction tuning leave the carriers in place: the same three heads, with full transfer. The last half of Pythia-410M pretraining behaves the same way. A finding about which heads carry IOI in a base model is therefore a finding about its midtrained, preference-tuned and instruction-tuned descendants, in line with the observation that fine-tuning enhances existing mechanisms (Prakash et al., 2024).

Before that, during pretraining, the carriers move (Figure 6). In Pythia-410M, IOI appears at step 3000; head 11.4 becomes the strongest carrier at step 2000 and stays so, while the second and third places change hands until step 66k, after which the

	attention role	ablation map	top-3 carriers	transfer	main carrier
Pretraining end → midtraining, 5B tokens	0.96	0.85	2/3	0.93	14.4
Pretraining end → midtraining mix 1, 51B	0.96	0.84	2/3	0.94	14.4
Pretraining end → midtraining mix 2, 51B	0.96	0.85	2/3	0.95	14.4
Pretraining end → midtraining mix 3, 51B	0.96	0.88	2/3	0.94	14.4
SFT → DPO	0.99	1.00	3/3	1.00	14.4
DPO → Instruct	1.00	1.00	3/3	1.00	14.4
Pythia-410M step 66k → 143k	0.92	0.78	3/3	0.99	11.4
Pythia-410M vs. its deduplicated sibling	0.47	0.15	0/3	0.09	12.12

Table 8: IOI carriers along a training lineage (OLMo-2-1B; Pythia-410M) and between the 410M siblings. Attention role: END→IO map; ablation map: reliability-corrected; transfer: damage of the earlier model’s top three in the later model, relative to its own top three; main carrier: the later model’s strongest head.

final top three are in place. Its deduplicated sibling settles on a different main carrier, 12.12, by step 6k and completes its final top three at step 100k. At 160M the two siblings take the same main carrier, 8.9, at step 3000, keep it to the end, and transfer 69–99% of their damage to each other from then on, which is why the 160M pair shares its carriers in Table 5. From the moment carriers appear, the two siblings share none of their top three, while their attention roles start at a correspondence of 0.79 and diverge slowly to 0.47. In DataDecide-1B, the head that carries IOI when the behaviour first appears is still the main carrier at the end of training in three of twelve runs; in the other nine, another head takes over later in pretraining.

Carrier assignment is thus an outcome of a run’s own training. Siblings begin from the same coordinate system, candidate heads compete for the task while the corpus is being learned, and the winner is kept once pretraining is over. This explains the two halves of Figure 4: siblings have each been through their own competition, descendants inherit its result.

6 Data interventions and the competition among candidates

The competition among candidates also determines what a training-data intervention does to a single head. Training-data attribution identifies the data that create a component by removing data, retraining, and reading out the same head (Chen et al., 2026). We run such interventions with controlled pretraining, where the initialization and the data order are fixed and only the intervention differs (Figure 7). From three initializations we mask the loss of 10% of the sequences during steps 500–2000, the window in which induction heads emerge: either the most repetitive sequences, the data that

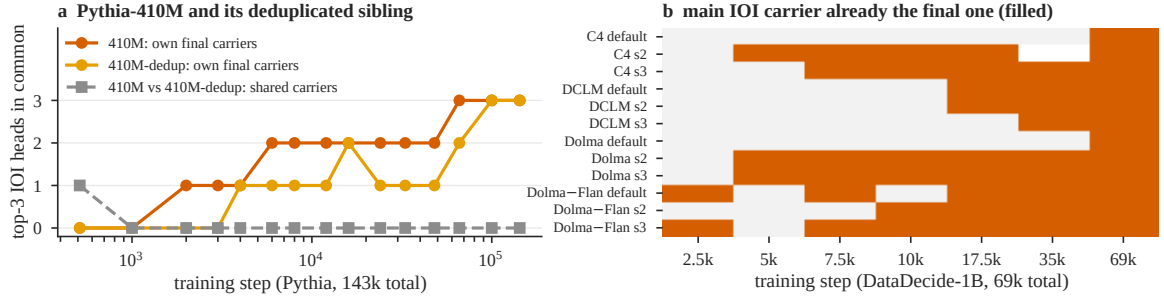


Figure 6: Carriers along pretraining. (a) Number of each Pythia-410M model’s final top-three IOI carriers already among its top three (solid), and number of top-three carriers shared by the two siblings (dashed). (b) Twelve DataDecide-1B runs (four corpora $\times$ three initializations): filled where the current main IOI carrier is the run’s final main carrier.

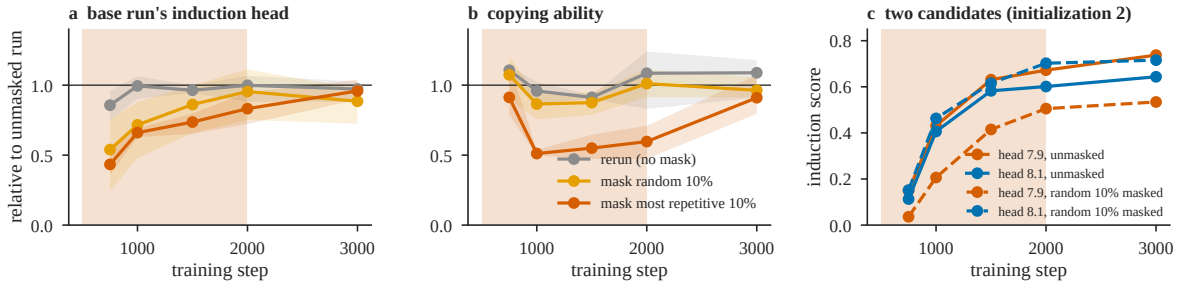


Figure 7: Masking training data while induction heads emerge (shaded: steps 500–2000) in controlled 12-layer models from three initializations. (a) Induction score of the unmasked run’s strongest induction head, relative to the unmasked run (mean and range over initializations); (b) copying ability (loss reduction on the repeated half of random sequences), relative. (c) Initialization 2 has two near-equal candidates; masking a random 10% of sequences hands the role to the second one at emergence, and it keeps it.

most accelerate induction heads (Chen et al., 2026), or a random 10%.

When one head leads the role, the intervention delays that head. Masking the most repetitive data lowers the strongest head’s induction score to 36–55% of the unmasked run at step 750 and copying ability to about half at step 1000; afterwards both recover, and at step 3000 the strongest induction head is the same head in all three initializations. A rerun of the unmasked configuration reproduces the same head in all three. Here a drop in the head’s score is a delay of the mechanism, and a single-head readout measures exactly that.

When two heads compete, a small change in the data decides the winner. In initialization 2, heads 7.9 and 8.1 rise together and 7.9 stays slightly ahead in the unmasked run and in its rerun. Masking a random 10% of the sequences lets 8.1 lead from the moment induction appears, and it keeps the role: at step 3000 the original head scores 72% of the unmasked run while the strongest induction

score (97%) and copying ability (92%) are intact. A readout of 7.9 alone would report a weakened mechanism where the mechanism has moved. In the two initializations with a clear leader, the same random masking leaves the leading head in place (95% and 98% of its unmasked score at step 3000) and copying ability intact (91% and 106%). Induction heads thus show, in miniature and under full control, the pattern of the IOI carriers: a role with one leading candidate stays on its head, and a role with several candidates goes to whichever candidate the run’s data favour when the role is formed.

7 Related Work

Universality and stability of components. Circuits are consistent in their algorithm across training and scale while components change (Tigges et al., 2024); attention heads are unstable across seeds, most in middle layers (Bali et al., 2026); positional consistency across Pythia seeds is lim-

ited (Trott, 2025); and representations of independently trained networks share structure up to rotation or permutation (Li et al., 2016; Kornblith et al., 2019; Ainsworth et al., 2023). Seed variants such as PolyPythias (van der Wal et al., 2025) and Multi-BERTs (Sellam et al., 2022) change initialization and data order together. We separate initialization from data, and separate attention roles from causal carriers, which explains why components agree among siblings in one sense and not in the other.

Circuit discovery. Circuits are found by activation patching and its automated and gradient-based variants (Wang et al., 2023; Conmy et al., 2023), and universality has been studied for small algorithmic models and for neurons (Chughtai et al., 2023; Gurnee et al., 2024). Our results say which of a discovered circuit’s statements to reuse in a related model and which to recompute, and that the recomputation needs a causal readout.

Mechanisms after further training. Fine-tuning reuses and enhances the circuits of the base model (Prakash et al., 2024). We show this for a full public lineage, through midtraining on new mixtures, SFT, DPO and instruction tuning, and contrast it with siblings trained separately.

Training data and components. Influence-based attribution traces heads to training samples and verifies the attribution by retraining (Chen et al., 2026); data properties such as burstiness shape when in-context learning emerges (Chan et al., 2022). Our controlled masking shows when such interventions delay the same head and when they move the role to another candidate.

Reliability of circuit measurements. Circuit scores vary across inputs and methods (Méloux et al., 2026), and redundant heads repair ablations (McGrath et al., 2023). Split-half reliability of every head map, and its correction of cross-model correspondence, make comparisons of causal maps between models interpretable.

8 Discussion

When to carry a finding over. Head-level findings carry over along a training lineage, from a checkpoint to its continued pretraining, mid-training, SFT, DPO and instruction-tuned descendants. Between siblings that share an initialization, attention-role labels carry over and a task’s carriers

are localized again; between independently initialized models, the layer and the algorithm carry over.

A protocol for reusing a circuit. To apply a circuit found in model A to model B: (i) establish the lineage from the weights, since released labels can be wrong (Appendix A); (ii) if B continues A’s training, reuse the named heads and confirm them with a single ablation of the named set; (iii) if B shares A’s initialization, compare attention-role maps head by head to inherit role labels, and localize the task’s carriers with single-head ablations in B; (iv) otherwise, transfer the algorithm and the layer profile and run circuit discovery in B; (v) in every case, report the split-half reliability of the head maps that are compared.

Consequences for data attribution and model diffing. Studies that retrain a model under a data intervention and track a named head compare two siblings that share the initialization and the data order. Our controlled runs show that such a comparison reads the mechanism correctly when one head leads the role, and that a role with close candidates can move under an intervention as small as a random 10% mask, so checking the strongest head of the role and the behaviour alongside the named head separates a delayed mechanism from a moved one. Comparisons between a base model and its fine-tuned or instruction-tuned descendants, as in model diffing, sit on the other side of the rule: the descendants keep the base model’s carriers, so differences between them can be attributed to the same named heads.

What the coordinate system is. The initialization does not choose the carriers; it chooses where each role can live. Heads that share a role’s attention profile are candidates for the task, and the run’s own data decide which candidate wins. The two are set at different times: the coordinates within the first few percent of training (Table 4), before the corpus has had much effect, and the carriers late in pretraining (Figure 6), after each run’s corpus has shaped the competition; this is why the coordinate system tracks the similarity of the corpora and the carriers do not. This places the variation that Tigges et al. (2024) and Bali et al. (2026) observe across seeds at a definite point: in the competition among candidates during pretraining, not in the coordinates of the roles, which the initialization sets, and not in later training, which keeps the outcome.

Limitations

We trace one task circuit, IOI, through causal carriers and nine roles through attention and weights; extending the carrier analysis to circuits for factual recall and in-context learning is the natural next step. The public lineages we use reach 12B parameters in Pythia and 1B in OLMo 2, and the controlled interventions use 12-layer models, so the same measurements on larger lineages with released midtraining and post-training stages would extend the rule to current frontier pipelines.

Ethical Considerations

This work analyses publicly released models and data. It reports lineage mismatches in public checkpoints so that users of these suites can compare the models they intend to compare.

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A Lineage of public checkpoints

Head layouts identify which checkpoints share an initialization, and we confirmed each case with the weights. (i) In DataDecide-1B, six runs were trained from initializations other than the one their seed label names; they form three pairs that share an unregistered initialization (step-0 weight correlation 0.000 with the labelled seed; step-2500 weights correlate within each pair as strongly as true siblings). We exclude them from all sibling analyses. (ii) The PolyPythias 160M weight-seed variants were trained from the standard initialization: their step-1 weights correlate 1.0 with the standard step 0, although their released step-0 files differ. (iii) The released final checkpoint of Pythia-2.8B-deduped is nearly identical to Pythia-2.8B (weight correlation 0.99), so this pair cannot serve as a same-initialization, different-data pair. (iv) The main revision of OLMo-2-0425-1B comes from a different training run than its released

stage-1 and stage-2 checkpoints and its SFT, DPO and Instruct models: a layer-5 query matrix correlates 0.000 with every stage checkpoint and with the Instruct model, while the Instruct model correlates 0.976 with the end of midtraining, and the head layouts agree accordingly. Lineage analyses should start from the stage-2 checkpoints.

B Role definitions

Induction: attention from the second copy of a repeated random block to the token after the earlier occurrence. Previous token, current token, two tokens back: attention to positions $t-1$, t , $t-2$ on natural text. Attention sink: attention to position 0. Knowledge retrieval: attention from the final prompt token to the answer entity in knowledge-conflict prompts. Duplicate token: attention from the second copy of a random block to the same token’s first occurrence. Delimiter: attention mass on punctuation and newline tokens. OV copying: $\sum \text{Re } \lambda / \sum |\lambda|$ over the eigenvalues of the full OV circuit. All attention roles are averaged over two halves of their inputs; split-half reliabilities are 0.98 or higher.

C Controlled pretraining

The models have 12 layers, 12 heads, width 768 and a 50k vocabulary, and are trained on C4 with sequences of 512 tokens, batches of 64 sequences, AdamW (learning rate 10^{-3} , 300 warm-up steps, cosine decay over 10k steps), for the first 3000 steps from three initializations with one data order. Masking sets the loss weight of the selected sequences to zero for steps 500–2000 and leaves the data order unchanged. A sequence’s repetition score is the share of tokens whose preceding bigram occurred earlier in the sequence followed by the same token; the masking threshold is the 90th percentile on C4 (0.08). Random masking selects 10% of sequences with a separate random generator.